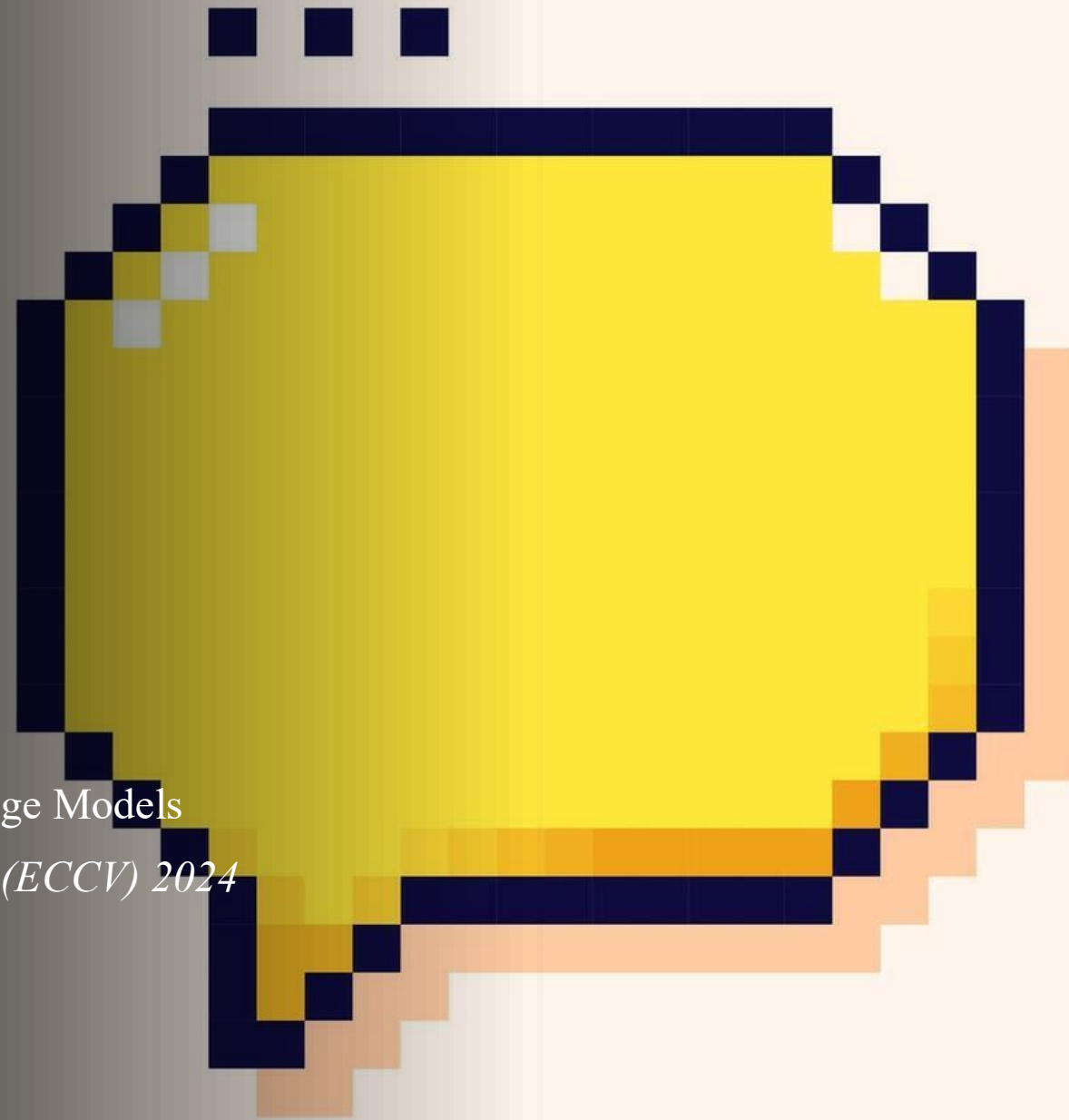


QPrompt

Quantized Prompt for Efficient Generalization of Vision-Language Models

Hao, Tianxiang, et al. *European Conference on Computer Vision(ECCV) 2024*

허정원 이진주 이찬영 최비결 안가영 황장훈



Contents

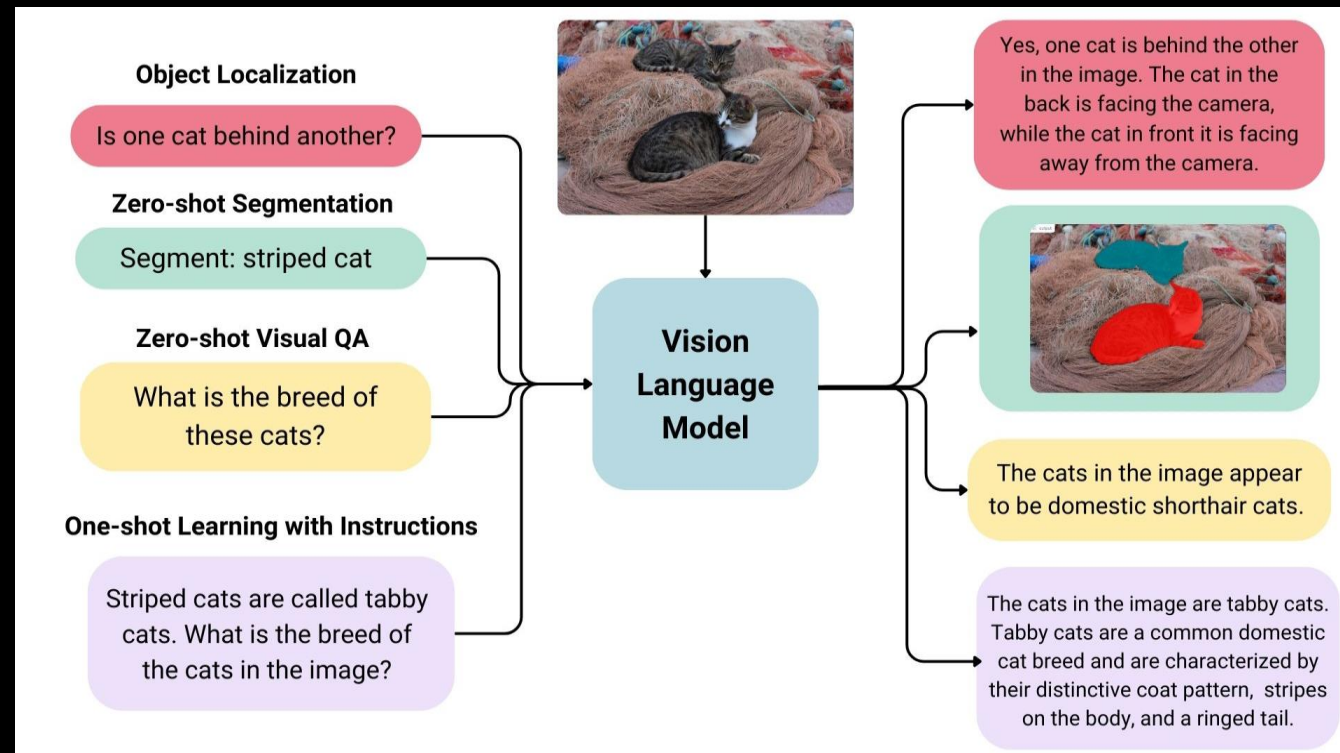
- **Introduction**
- **Related Work**
- **Exploring Quantization in Model Generalization**
- **Prompt Quantization**
- **Experiments**
- **Conclusion**

Contents

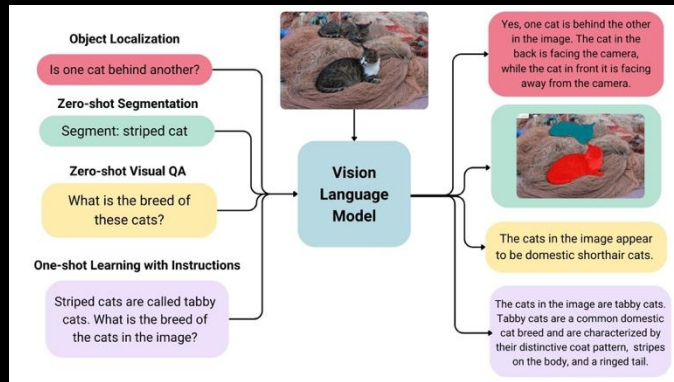
- **Introduction**
- Related Work
- Exploring Quantization in Model Generalization
- Prompt Quantization
- Experiments
- Conclusion

Introduction

- Large-scale pre-trained **vision-language model** have achieved tremendous success in various fields.



Introduction



Huge pre-trained models

MaPLE, CoOp

Overfitting

Catastrophic Forgetting

Downstream tasks and dataset

- How to **transfer** the rich knowledge in such huge pre-trained models to **downstream tasks and dataset**?

Introduction

- Analyze the effect of noise and rethink the relationship between **noise and generalization**.
- This work is the first to propose to **quantize prompts**.
- Reach competitive results, **winning many state-of-the-art tuning methods with a much smaller size of model**.

Contents

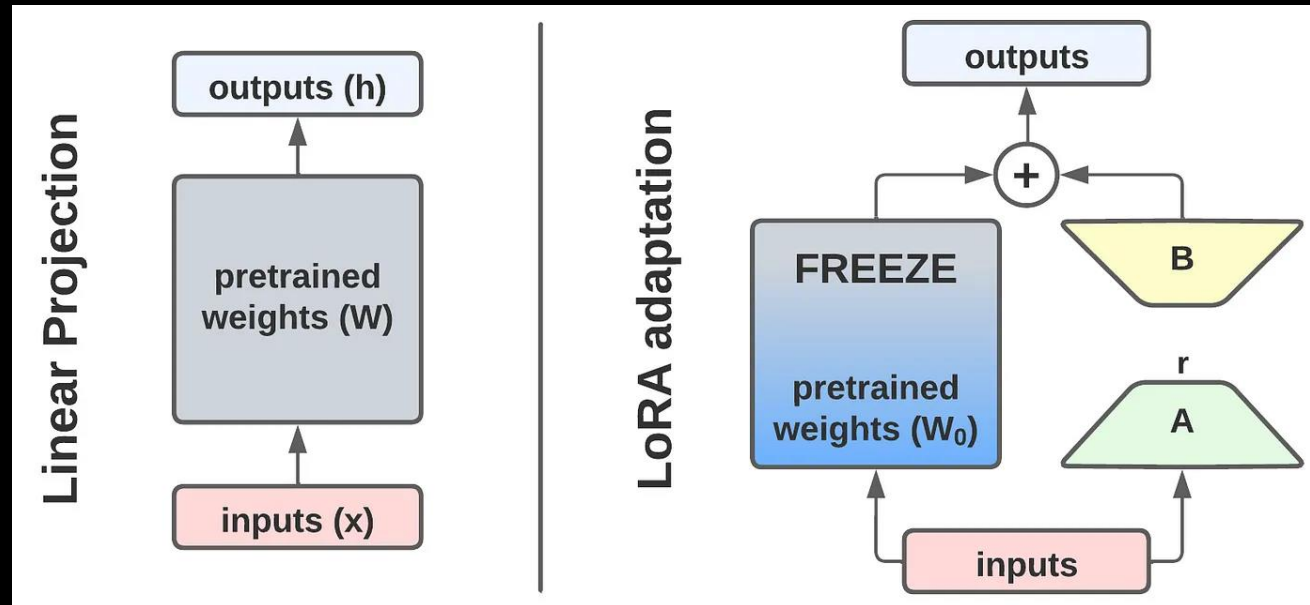
- Introduction
- **Related Work**
- Exploring Quantization in Model Generalization
- Prompt Quantization
- Experiments
- Conclusion

Related Work – Vision Language Model

- **Large-scale vision-language models** show strong performance in various tasks.
- Classic works learn multimodal representations via self-supervised learning from **large image-text pairs**.
- **CLIP** aligns vision and language with contrastive learning, achieving exceptional results.
- **Applications:** few-shot recognition, detection, segmentation, video classification, video understanding.

Related Work – PEFT

- **PEFT** transfers pretrained models to downstream tasks with minimal parameters.
 - Origin: NLP → adapter, prompt tuning
 - LoRA: freeze backbone, tune small extra modules.



Related Work – PEFT

- Extended to vision models and VLMs:
 - **CoOp**: replace CLIP fixed templates with tunable **text** prompts.
 - **CoCoOp**: use **image** features to guide prompt optimization.
 - **Others**: optimize both image/text prompts, add modality connections.
- Mitigate overfitting/forgetting via regularization modules or losses.

Related Work – Quantization

- Effective compression: reduce weights/activations from FP32 to lower bit-width (e.g., INT8).
 - lower memory footprint, reduced computation, minimal accuracy loss.

Two types:

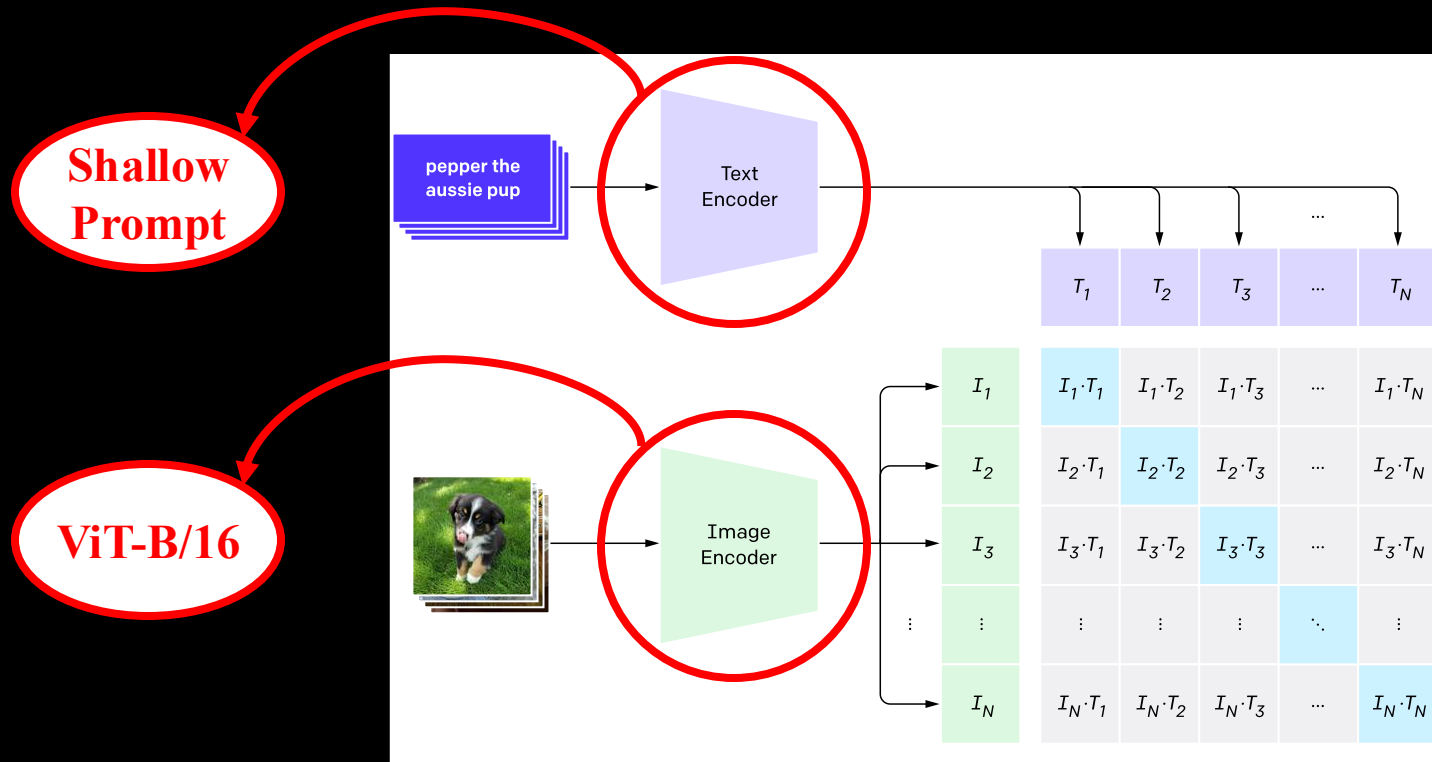
- PTQ: few resources, higher accuracy loss.
- QAT: more resources, better accuracy.
- Most works aim to minimize quantization error; this work shows both **excessive and insufficient error harm generalization** → **moderate error is optimal**.

Contents

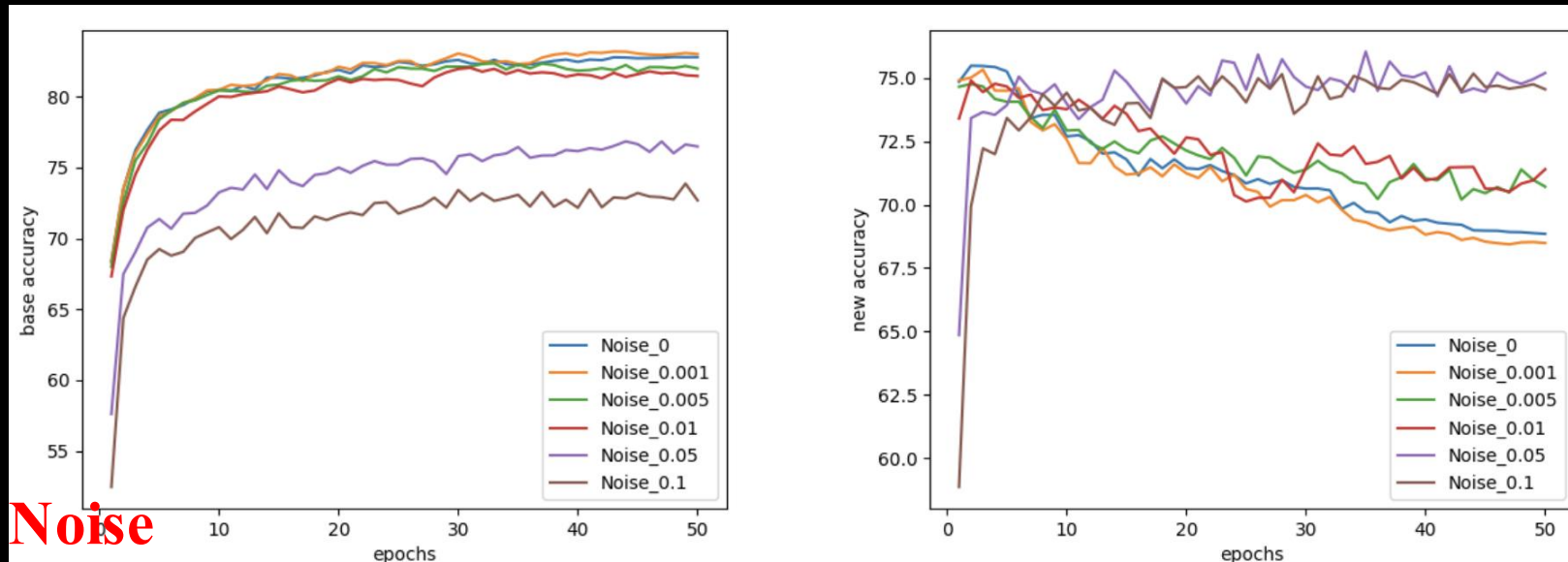
- Introduction
- Related Work
- **Exploring Quantization in Model Generalization**
- Prompt Quantization
- Experiments
- Conclusion

Prompt tuning of VLM

- CLIP comprises a text encoder $\mathcal{L}$ and an image encoder $\mathcal{V}$.



Rethinking the Relationship between Noise and Generalization



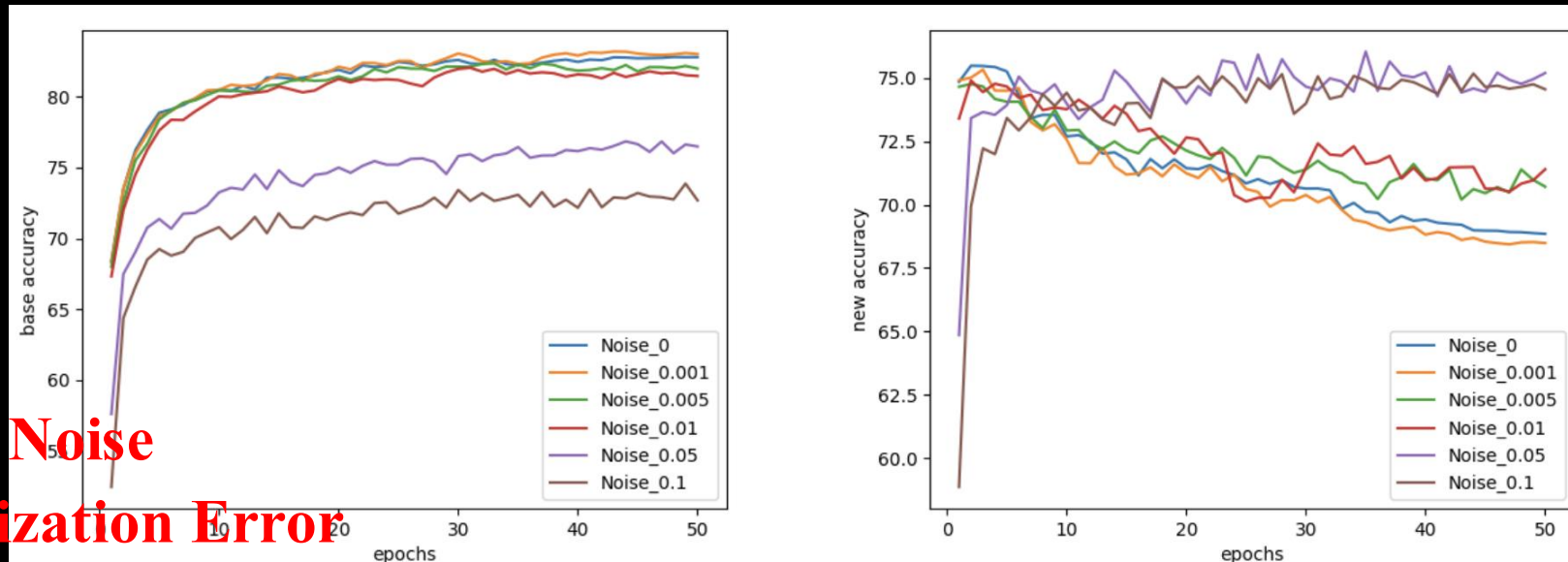
Gaussian Noise

- Only noise of moderate intensity outperforms baseline in **specialization-generalization** trade-off.
- **Quantization** can also be viewed as introducing some form of **noise** into the parameters, and thus can possibly **improve the generalizability as the noise**.

Rethinking the Relationship between Noise and Generalization

Gaussian Noise

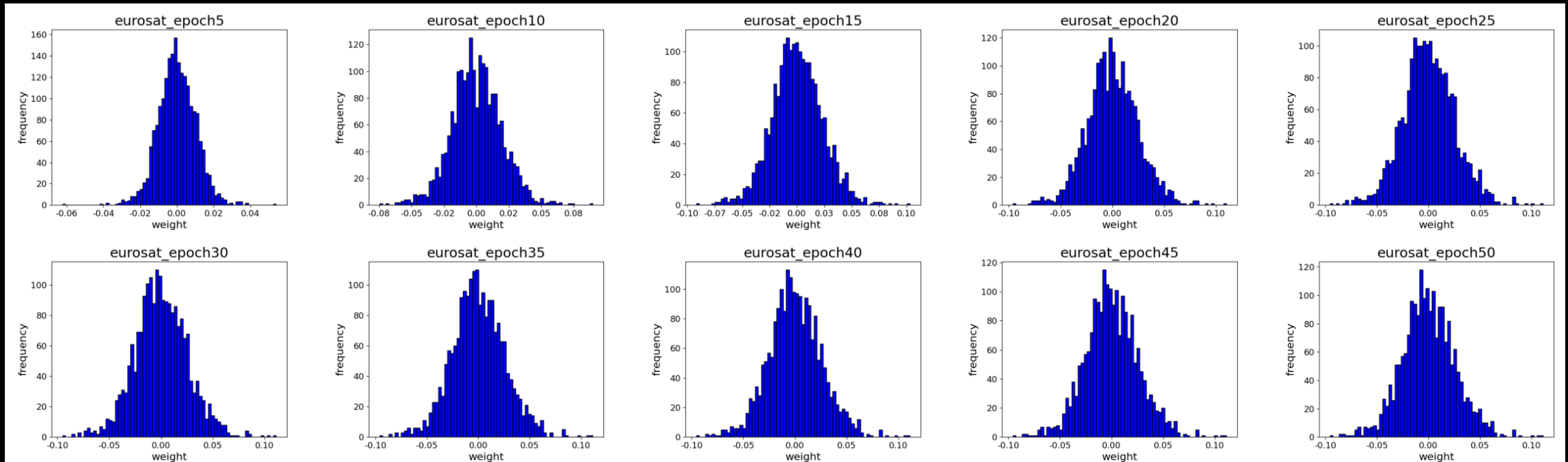
→ **Quantization Error**



Quantization error...

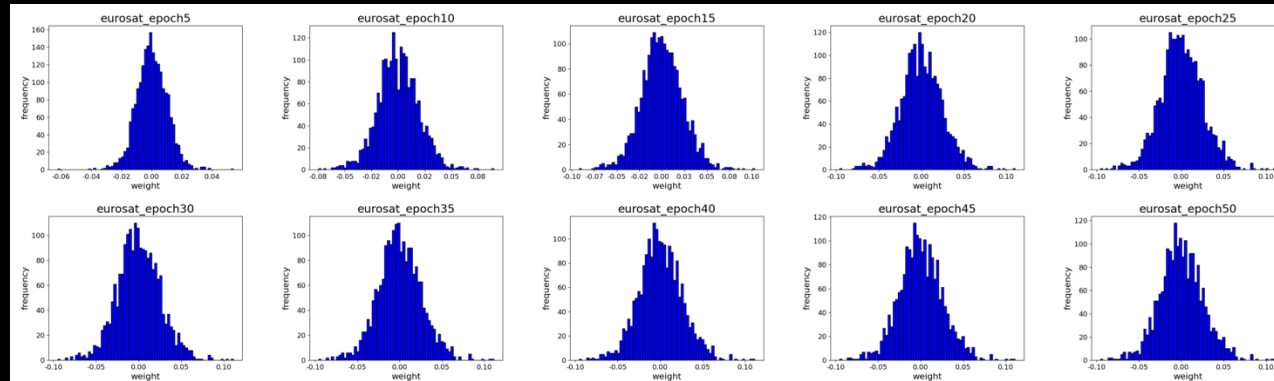
- is more controllable than gaussian noise
- can reduce the storage required

Characteristics of Prompts in Vision-Language Models



- The histogram about the frequency of weights of prompts during training of CoOp on eurosat dataset.

Characteristics of Prompts in Vision-Language Models



- Because prompt weights are **noise-tolerant**, aggressive quantization (e.g., 1-bit) can be applied without harming generalization.
- Since prompts are a **small fraction** of parameters, **QAT's extra cost is minimal**, and quantization error should be kept at a moderate level rather than minimized.
- Prompt weight distribution **shape remains stable**, so normalizing before quantization and denormalizing after may improve accuracy.
- With **almost no outliers**, simple clustering methods like **K-Means become viable** and efficient alternatives to complex parametric approaches.

Contents

- Introduction
- Related Work
- Exploring Quantization in Model Generalization
- **Prompt Quantization**
- Experiments
- Conclusion

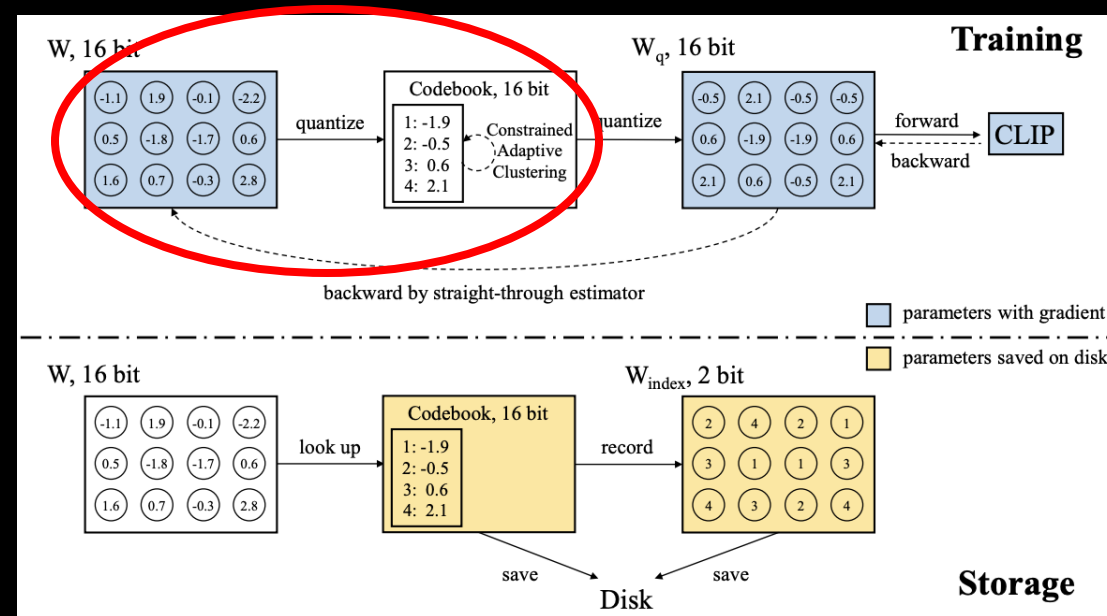
Quantization Basis

- Suppose there are m parameters in total to be quantized, which are denoted by $W \in R_m$. Each w_i here is a high-precision float-type number.
- A mapping Q from intervals to points, mapping all values within an interval U_i to the same quantized value q_i , i.e. $Q(x) = q_i, \forall x \in U_i$.
- The quantization error E :

$$E = \sum_{i=1}^N ||Q(W_i) - W_i||^2$$

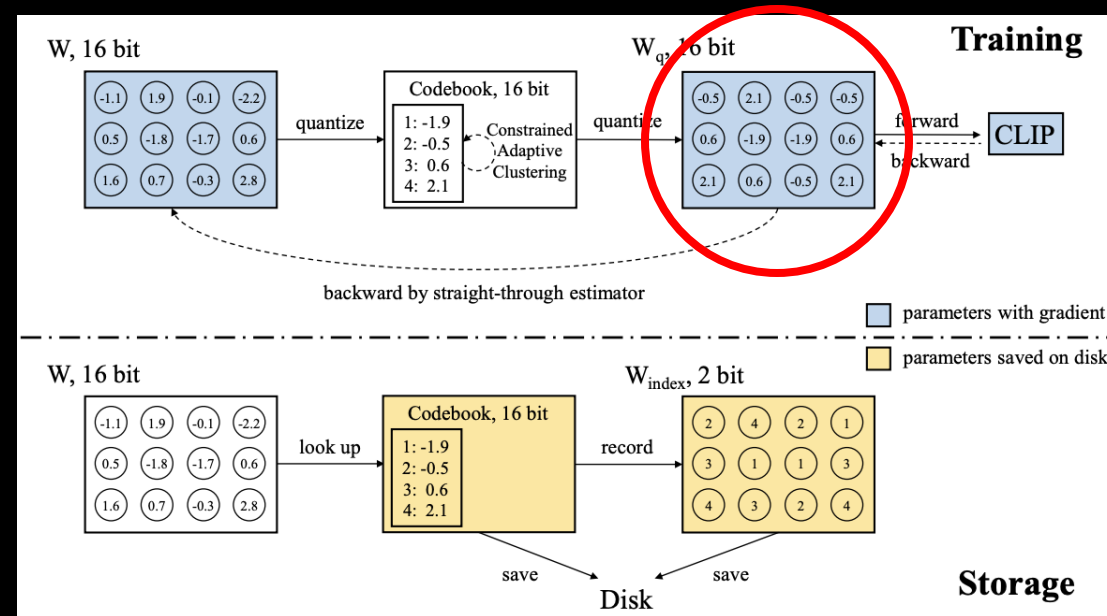
Such error serves as the objective of **K-Means algorithm** as well, so it can be ensured that each time clustering is **redone using the K-Means algorithm**, the **quantization error will most likely decrease**.

Overview of Quantized Prompt



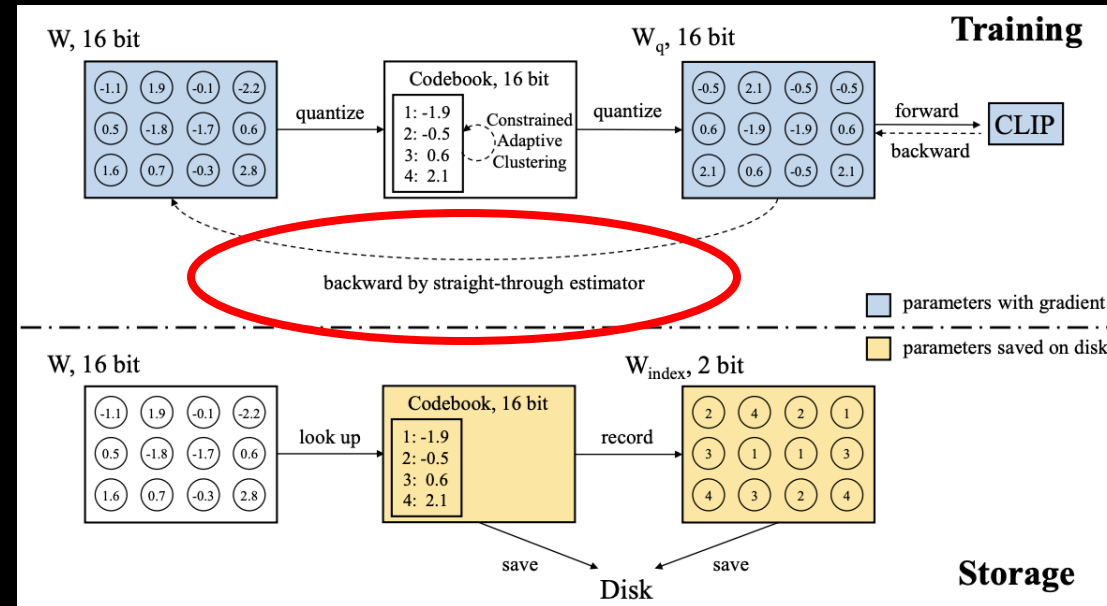
1. K-Means algorithm is run to fit the pre-trained prompt weights, and the cluster centers (codebook) are recorded.

Overview of Quantized Prompt



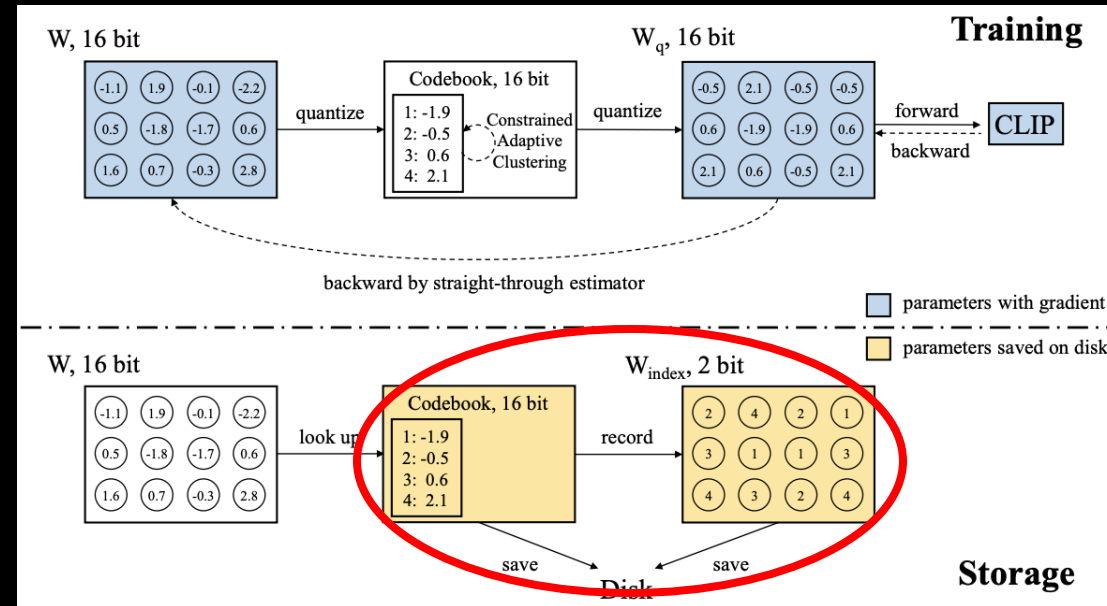
2. The prompt weights are normalized using mean and variance, apply K-Means quantization on the normalized weights, and then denormalize to obtain the quantized prompt.

Overview of Quantized Prompt



3. Since Q is non-differentiable, we use Straight-Through Estimator (STE) to propagate gradients, tuning weights by fake gradients during training.

Overview of Quantized Prompt



4. For storage, fp16 parameters are converted to b-bit indexes with a small codebook, reducing storage from $16N$ bits to $bN + 2^b \times 16$, achieving up to $16\times$ compression when $b = 1$.

Constrained Adaptive Clustering for Quantization

- Re-clustering the prompt every iteration like classic QAT algorithms did to keep the codebook updated at all times is not a good choice.
1. **K-Means algorithm is not quite efficient**, and thus if we run it every iteration then the training efficiency would decrease
 2. Only moderate noise promotes generalization, so keep updating the clusters to **minimize the quantization error may be not helpful** but harmful
 3. The weight changes are very gentle for most of the training time, and **the weight distributions of adjacent stages are highly similar**.

Constrained Adaptive Clustering for Quantization

- **Constrained Adaptive Clustering** is proposed to guide parameter updates in K-Means, which are not updated by gradient.
 - First, set a minimum cluster update interval t .
 - Second, perform re-clustering when current weight distribution is far different.
- To reach the goal, **Kullback-Leibler divergence** is a good metric.
 - To enable comparison, both distributions are projected into the same event space.

$$KL(p^{cur} || p^{old}) = \sum_{i=1}^{2^b} p^{cur}(i) \log \frac{p^{cur}(i)}{p^{old}(i)}$$

The update would continue only if Kullback-Leibler divergence exceed a certain threshold T_{KL}

Contents

- Introduction
- Related Work
- Exploring Quantization in Model Generalization
- Prompt Quantization
- **Experiments**
- Conclusion

Experiments

1. **Base-to-new generalization**, which models are trained with base classes and evaluated on **both base and new classes**.
2. **Domain generalization**, which aims to show generalization to the domain shift, especially for **out-of-distribution data**.
3. **Cross-dataset transfer**, which aims to see if the method has the potential to transfer beyond a single dataset. It is a much more challenging problem because the fundamentals can be **totally changed across different datasets**.
4. **Few-shot learning**, which aims to evaluate **the adaptation performance** of the model to extract knowledge from a dataset whose **samples are very few**, e.g. 1, 2, 4, 8 or 16 samples.

Experiments - Base-to-new Generalization

	Size	Base	New	H
CLIP [59]	0KB	69.34	74.22	71.70
CoOp [78]	4.1KB	82.69	63.22	71.66
CoCoOp [77]	70.8KB	80.47	71.69	75.83
Adapter [17]	1051KB	82.62	70.97	76.35
LoRA [25]	258KB	84.30	67.33	74.86
ProGrad [79]	16.4KB	82.79	68.55	75.00
QCoOp	0.26KB	80.68	74.44	77.43
MaPLe [33]	7096KB	82.28	75.14	78.55
QMaPLe	1774KB	83.02	75.57	79.12

- **Table 1** Comparisons with latest methods in base-to-new generalization. H: harmonic mean.

Experiments - Domain Generalization

	Size	Source ImageNet	Target				Average
			-V2	-Sketch	-Adversarial	-Rendition	
CLIP	0KB	66.73	60.83	46.15	47.77	73.96	57.18
CoOp	4.1KB	71.51	64.20	47.99	49.71	75.21	59.28
CoCoOp	70.8KB	71.02	64.07	48.75	50.63	76.18	59.91
Adapter	1051KB	69.33	62.53	47.67	49.17	75.42	58.70
LoRA	258KB	70.30	62.37	42.43	38.40	68.97	53.04
ProGrad	16.4KB	72.24	64.73	47.61	49.39	74.58	59.07
QCoOp	0.26KB	70.67	63.87	48.93	51.10	76.90	60.20

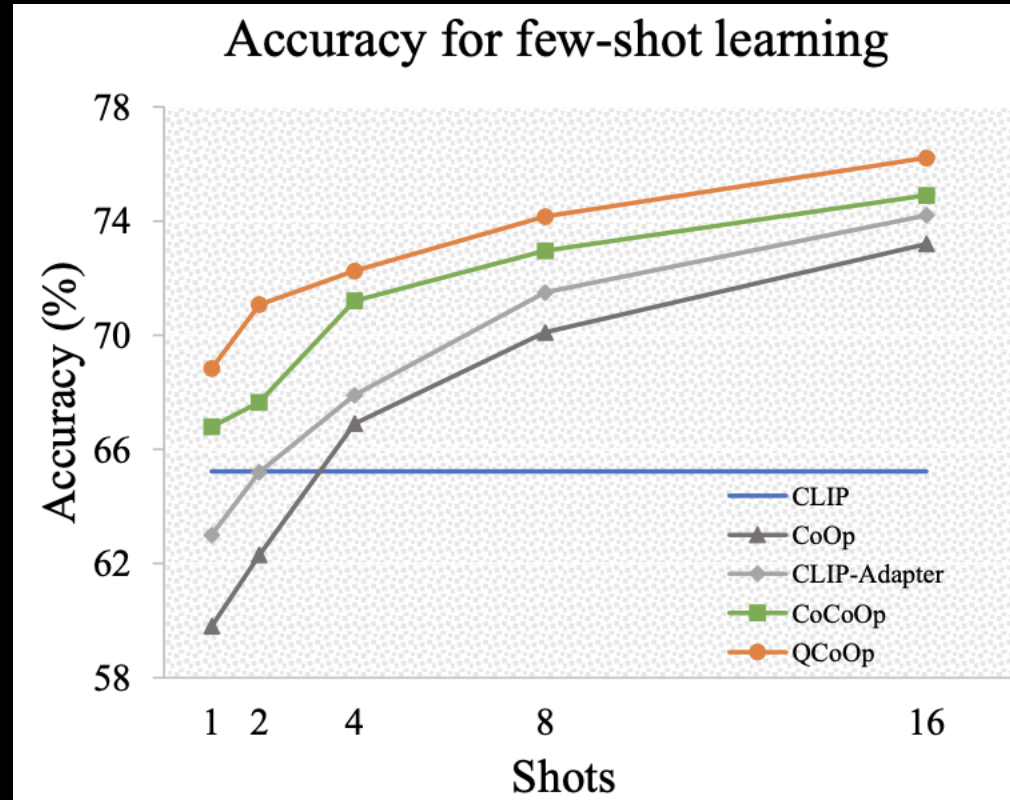
- **Table 2** Comparisons with latest methods in domain generalization.

Experiments - Cross-dataset Transfer

		Source	Target										
Size		ImageNet	Caltech101	Pets	Cars	Flowers	Food101	Aircraft	Sun397	DTD	EuroSAT	UCF101	Average
CoOp	4.1KB	71.51	93.70	89.14	64.51	68.71	85.30	18.47	64.15	41.92	46.39	66.55	63.88
CoCoOp	70.8KB	71.02	94.43	90.14	65.32	71.88	86.06	22.94	67.36	45.73	45.37	68.21	65.74
Adapter	1051KB	69.33	93.43	88.87	64.40	70.27	85.63	24.67	65.80	44.90	47.70	66.00	65.17
QCoOp	0.26KB	70.63	94.07	90.53	65.97	71.33	86.23	22.73	66.80	44.20	48.23	69.17	65.93

- **Table 3** Results in the cross-dataset transfer setting.

Experiments - Few-shot Learning



- **Table 5.** Average few-shot learning results on 11 datasets.

Contents

- Introduction
- Related Work
- Exploring Quantization in Model Generalization
- Prompt Quantization
- Experiments
- **Conclusion**

Conclusion

- **Effectively and efficiently adapting such huge models to downstream tasks** is a challenging problem.
- Use **K-Means** clustering algorithm as the base of our quantization method.
- Propose a dynamic check(**Kullback-Leibler divergence**) every few iterations.
- The proposed method can be **easily integrated** into many existing vision-language tuning methods, such as CoOp and MaPLe, while achieving good performance.

Questions?